ECHO CHAMBER BREAKER

Complete Rigorous Logic System

Decision Architecture • Stance Gating • Scoring • Tool Contracts • Edge Cases

Document Type	Normative Logic Specification (source of truth for implementation)
Audience	Developers, Antigravity agents, evaluators
Scope	All decision points from input reception to honesty score mutation
Principle	No tool may invent a conviction the user did not express
Version	1.0 — aligned with Product Definition + Technical Architecture

This document defines the complete logical behavior of Echo Chamber Breaker. Every implementation decision (frontend gate, tool handler, scoring engine, UI state machine) must be traceable to a rule in this specification. Ambiguity is resolved in favor of not inventing stances the user did not express.

1. Core Axioms (Non-Negotiable)

A1	**No invented conviction**	The system must never treat a greeting, question, incomplete fragment, or neutral statement as a strong opinion that requires steelmanning.
A2	**Steelman only real stances**	Steelmanning is only permitted after a positive stance classification. Weak or sarcastic counters are forbidden.
A3	**Score only on engagement actions**	The Intellectual Honesty Score changes only when the user takes an explicit engagement action (Dismiss, Concede, Submit Rebuttal, Complete Challenge, Write Reflection). Viewing is free.
A4	**Explainable score**	Every score mutation is an append-only ledger event with reason + delta. Current score is derived, never stored as the sole source of truth.
A5	**Voice-first output contract**	Every tool response that may be spoken must include a short 'say' field (≤ ~40 words) optimized for Alexa synthesis, plus structured content for screens.
A6	**No moralizing**	The system challenges the strength of reasoning and engagement quality. It does not assign moral worth, political correctness scores, or ideology labels.
A7	**Authentic MCP**	The simulator (and any client) must call the real MCP endpoint. Frontend must not invent tool results.

2. Input Classification (Stance Gate)

Every user utterance first passes through a Stance Gate. This is the most important logic layer. Failure here produces the 'hi Alexa → STRONG_AGREE 65%' failure mode observed in early builds.

2.1 Classification Categories

Category	Definition	System Behavior
GREETING / SOCIAL	Hello, hi, thanks, how are you, small talk, pure social	Light reply. Offer to receive a conviction. No analysis.
QUESTION	Interrogative form seeking information, not asserting a view	Answer briefly or invite the user to state a position they hold.
INCOMPLETE / FRAGMENT	Too short, truncated, or lacking evaluative content	Ask for a fuller statement of the conviction.
NEUTRAL / DESCRIPTIVE	Factual description without clear evaluative stance	Ask whether the user wants this treated as a position to challenge.
STANCE / CONVICTION	Clear opinion, belief, moral claim, preference, prediction with confidence	Proceed to full analyze_stance → diagnosis → steelman offer.
META / SYSTEM	Commands about the product itself, score, history, export	Route to the appropriate system tool (honesty_report, etc.).

2.2 Stance Detection Signals (Positive Evidence)

- Evaluative language: good/bad, should/must, always/never, best/worst, right/wrong
- Strong modality: clearly, obviously, definitely, everyone knows, no one can deny
- Identity-linked claims: 'I believe', 'I feel that', 'in my view', 'as a … I know'
- Moral or policy prescriptions: 'ought to', 'must ban', 'should never'
- Comparative superiority claims about groups, systems, technologies, cultures
- Length and structure consistent with an argument rather than a query

2.3 Negative / Blocking Signals

- Starts with greeting tokens (hi, hello, hey, good morning) and contains little else
- Pure interrogative with no embedded assertion
- Single word or under ~8–10 tokens without evaluative content
- Requests for definition, explanation, or how-to without a position
- Explicit meta: 'what is my score', 'show history', 'challenge me'

2.4 Decision Rule (Normative)

```
IF input matches META/SYSTEM → route to system tool
ELSE IF input is GREETING | QUESTION | INCOMPLETE | NEUTRAL:
    return { is_stance: false, reason: "...", say: soft invitation }
ELSE IF input is STANCE/CONVICTION:
    proceed to analyze_stance (full pipeline)
ELSE:
    treat as ambiguous → ask clarifying question
    (do NOT invent a stance)
```

3. analyze_stance Tool Contract

This tool is only called after the Stance Gate has classified the input as STANCE/CONVICTION (or the user has confirmed an ambiguous input). It never invents a stance.

3.1 Required Inputs

- statement (string) — the exact user text
- userId (string) — for session / history context
- optional: topicHint, previousChallengeId

3.2 Required Output Schema

```
{
  "is_stance": true,
  "polarity": "STRONG_AGREE" | "AGREE" | "LEAN" | "NEUTRAL" | "DISAGREE" | "STRONG_DISAGREE",
  "conviction_certainty": 0-100,          // calibrated confidence, not raw model certainty
  "emotional_charge": 0-100,              // affect intensity
  "blind_spots": [                        // 2–4 specific, non-generic items
    "concrete missing consideration 1",
    "..."
  ],
  "unexamined_assumption": "one crisp sentence",
  "topic_label": "short label for UI",
  "say": "≤40 word spoken summary for Alexa",
  "structuredContent": { ... full diagnosis object ... }
}
```

3.3 Quality Rules for Blind Spots

- Must be specific to the actual claim, not generic templates.
- Must be charitable: name considerations a thoughtful opponent would raise.
- Must not moralize or attack the user's character.
- Prefer systemic incentives, historical counter-examples, secondary effects, and definitional ambiguity.

4. steelman Tool Contract

Steelmanning is the core product differentiator. A weak or sarcastic counter destroys credibility. This tool may only be invoked after a successful analyze_stance with is_stance=true.

4.1 Steelmanning Rules (Hard)

R1	**Charity first**	Present the strongest reasonable version of the opposing view, not the weakest.
R2	**No strawman**	Do not caricature, mock, or reduce the original claim to an absurdity.
R3	**No agreement theater**	Do not weakly agree with the user. The point is productive friction.
R4	**Evidence-oriented**	Prefer structural reasons, incentives, historical patterns, and trade-offs over slogans.
R5	**Voice constraint**	The spoken 'say' version must be ≤ ~40 words and still carry the core counter-thesis.
R6	**Separate layers**	Return both a short spoken synthesis and a fuller foundational counter-thesis + premises.

4.2 Output Structure

```
{
  "say": "short Alexa-ready counter (≤40 words)",
  "foundational_counter_thesis": "one clear opposing thesis sentence",
  "core_structural_premises": ["premise 1", "premise 2", "premise 3"],
  "socratic_inquiry": "one sharp question that exposes a tension",
  "challenge_id": "uuid",
  "structuredContent": { ... }
}
```

4.3 Forbidden Behaviors

- Generating a remote-work steelman when the user talked about something else (content mismatch).
- Using the currently selected topic chip instead of the actual statement text.
- Producing generic 'there are two sides' platitudes without a real opposing thesis.
- Attacking the user's identity, ethnicity, religion, or group membership.
- Claiming the counter-view is the 'correct' absolute truth.

5. Engagement Actions & Scoring Logic

The Intellectual Honesty Score is a private longitudinal signal of engagement quality. It is not a moral grade and must not punish disagreement itself.

5.1 Canonical Engagement Actions

Action	Meaning	Default Δ	Notes
Dismiss	Rejects the steelman without engagement	–3.0	Visible cost, not punitive lecture
Concede Valid Point	Explicitly acknowledges a strong opposing consideration	+8.5	Highest single reward
Submit Counter-Rebuttal	Engages by arguing back against the steelman	+4.5	Quality of rebuttal can modulate later
Complete Multi-Turn Debate	Finishes a dialectical session	+6.0 to +12	Depends on depth & concessions
Write Reflection	User writes a short reflection after a challenge	+5.0	Optional but high-signal
Ignore / View Only	User sees the steelman but takes no action	0	Viewing is free (Axiom A3)

5.2 Score Derivation Rules

- Baseline on first use: 50.0 (neutral).
- Score = 50 + sum of all ledger deltas (clamped to a sensible range, e.g. 0–100 for display).
- Every mutation creates an append-only HonestyEvent: {eventId, delta, reason, relatedChallengeId, timestamp}.
- UI labels (CONFIRMATION_TRAPPED, CALIBRATING, OPEN_MINDED, etc.) are derived from score bands + recent pattern, not stored as facts.
- Repeated pure Dismiss actions without any positive engagement should produce a clear downward trend and a diagnostic label, not a lecture.

5.3 What Must Never Move the Score

- Simply opening the app or viewing the ledger.
- Receiving a steelman without choosing an action.
- The content of the user’s original opinion (agreeing or disagreeing with the system is irrelevant).
- Topic selection chips or example statements.
- System errors or failed tool calls.

6. Multi-Turn Dialectical Session Logic

6.1 Session Lifecycle

```
start_debate(challenge_id | statement)
    → create DebateSession { sessionId, turns: [], status: "active", maxTurns }
continue_debate(sessionId, user_turn_text)
    → append user turn
    → generate system counter-turn (steelman continuation)
    → append system turn
    → optionally update interim quality signals
conclude_debate(sessionId)
    → status = "concluded"
    → compute final engagement quality
    → emit HonestyEvent(s)
    → return summary + score impact
```

6.2 Turn Quality Heuristics (for later modulation)

- Length + presence of reasons (not pure assertion).
- Acknowledgement of opposing premises before rebutting.
- Introduction of new considerations vs pure repetition.
- Explicit concession language ('fair point', 'I hadn't considered').

6.3 Hard Constraints

- A debate session is always anchored to a specific original statement / challenge_id.
- System turns must remain steelman-quality; they may not degrade into debate-winning rhetoric.
- Max turns should be enforced (e.g. 4–6 total turns) to keep voice sessions usable.
- Concluding without any user engagement after the first steelman is treated as low-effort (small or zero positive delta).

7. Client / Server State Machine

7.1 Primary UI States

State	Description	Allowed Transitions
IDLE	Waiting for input. No active challenge.	→ CLASSIFYING on submit
CLASSIFYING	Stance Gate running	→ NON_STANCE_REPLY | ANALYZING
NON_STANCE_REPLY	Soft invitation / clarification shown	→ IDLE
ANALYZING	analyze_stance in flight	→ DIAGNOSIS_SHOWN | ERROR
DIAGNOSIS_SHOWN	Stance diagnosis visible; steelman not yet requested	→ STEELMANNING | IDLE
STEELMANNING	steelman tool in flight	→ CHALLENGE_ACTIVE | ERROR
CHALLENGE_ACTIVE	Steelman shown; user can Dismiss / Concede / Rebut / Debate	→ ENGAGING | DEBATING | IDLE
ENGAGING	record_engagement in flight	→ SCORE_UPDATED
DEBATING	Multi-turn session active	→ DEBATING | CONCLUDING
SCORE_UPDATED	Ledger event applied; UI reflects new score	→ IDLE | CHALLENGE_ACTIVE

7.2 Invariant

The UI must never display a Cognitive Stance Diagnosis or Steelman for an input that the Stance Gate classified as non-stance. Any residual diagnosis panel from a previous interaction must be cleared when a new non-stance input is processed.

8. Edge Cases & Known Failure Modes

Case	Required Behavior
Greeting only (“hi Alexa”)	NON_STANCE_REPLY. Soft invitation. No diagnosis, no score change.
Extremely short fragment	Ask for fuller statement. Do not invent polarity or certainty.
Topic chip selected + different text typed	Always analyze the text in the box. Chip is only a pre-fill helper.
Inflammatory / hostile content	Still steelman charitably if it is a real stance. Do not moralize or refuse on political grounds. Do not attack identity groups.
User dismisses repeatedly	Apply −3 each time. Update label (e.g. CONFIRMATION_TRAPPED). No lectures.
Bedrock timeout / error	Surface a clear error. Do not fall back to a previous unrelated steelman. Score unchanged.
User asks for their score	Route to honesty_report. Do not treat as a stance.
Empty or whitespace-only input	Ignore or prompt for input. No tool call.
User pastes a long multi-paragraph manifesto	Still classify as stance if evaluative. Steelman the core thesis; do not try to refute every sentence.
Simulator disconnected from MCP	Show clear “MCP offline” state. Never invent tool results in the UI.

9. Canonical Tool Call Sequences

9.1 Happy Path — Single Challenge

```
1. [Client] Stance Gate (local or via analyze_stance with is_stance check)
2. tools/call analyze_stance     → diagnosis
3. tools/call steelman           → counter-case + challenge_id
4. [User chooses action]
5. tools/call record_engagement  → HonestyEvent + new score
6. (optional) tools/call honesty_report
```

9.2 Happy Path — Multi-Turn

```
1–3. same as above
4. tools/call start_debate(challenge_id)
5. loop: tools/call continue_debate(sessionId, user_text)
6. tools/call conclude_debate(sessionId) → final events + score
```

9.3 Proactive Path

```
1. tools/call challenge_me(userId)
2. System selects a prior topic or weak spot from history
3. Present steelman-style challenge
4. Same engagement → record_engagement flow
```

9.4 Dependency Rules

- steelman requires a prior successful analyze_stance (or equivalent stance object) for the same statement.
- record_engagement requires a valid challenge_id that exists in the store.
- continue_debate requires an active sessionId.
- honesty_report may be called at any time; it is read-only.
- challenge_me may be called at any time; it should respect recent engagement fatigue (do not spam).

10. Implementation Acceptance Checklist

A build is considered logic-correct only when all of the following pass:

- ■ Greeting “hi Alexa” produces NON_STANCE_REPLY and never a STRONG_AGREE diagnosis.
- ■ A clear opinion statement produces is_stance=true and a specific, non-generic diagnosis.
- ■ Steelman always operates on the exact text currently submitted, never on a stale topic chip.
- ■ Dismiss / Concede / Rebuttal are the only user actions that move the score (viewing does not).
- ■ Every score change appears as an append-only ledger event with reason + delta.
- ■ Live MCP Protocol Trace shows real JSON-RPC traffic; UI does not invent results.
- ■ Every tool response that may be spoken includes a ≤40-word 'say' field.
- ■ Multi-turn debate stays anchored to the original challenge and maintains steelman quality.
- ■ Error / timeout paths never display a previous unrelated steelman.
- ■ Topic chips only pre-fill; they never override the analysis target.

This Logic System is the normative reference. When implementation and this document conflict, this document wins until it is deliberately revised. The central invariant remains: the system must never invent a conviction the user did not express, and must never move the honesty score without a real engagement action.

Echo Chamber Breaker — Logic before spectacle.